

HOWTO login

From HPC

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SSH connection details

HPC1 (also known as Rhasatsha)

Use the following connection details to log in:

- **Host Name:** hpc1.sun.ac.za
- **Port:** 22 (the default)
- **Connection type:** SSH (the default)

Please note that your username and password are the same as for your email.

HPC2 (also known as CAF-HPC1)

Use the following connection details to log in:

- **Host Name:** hpc2.sun.ac.za
- **Port:** 22 (the default)
- **Connection type:** SSH (the default)

Please note that your username and password are the same as for your email.

VPN ACCESS OFF CAMPUS

You will only be able to connect to the HPC while on campus (wired only, **no Wifi access**), or if you have a VPN connection open to campus. There are two VPN software options; the Citrix VPN client or PaloAlto GlobalProtect.

Citrix VPN. Download it from <https://securevpn.sun.ac.za> and run it before trying to ssh to the HPC.

GlobalProtect Read the HowTo on <https://servicedesk.sun.ac.za/kb/display/ITHUB/Global+Protect+VPN>

When on campus, do NOT use the VPN because you are connected to the real Stellenbosch network and do not need the Virtual adapter.

SSH clients

All interaction with the cluster is done securely with SSH.

Your interaction with the cluster will entail two major activities:

- file transfer
- job submission

Linux / macOS clients

Use the Terminal or Konsole to SSH to the cluster using the `ssh` command in the form:

```
$ ssh username@hpc1.sun.ac.za
```

```
$ ssh username@hpc2.sun.ac.za
```

To transfer files, use Konqueror (if you are using KDE) and the `fish` command in the form:

```
fish://username@hpc1.sun.ac.za
```

```
fish://username@hpc2.sun.ac.za
```

If you are using Gnome, open Gnome Commander and create a new *Remote Connection*.

You can also use the Terminal or Konsole to transfer files using the `scp` command.

To copy a file **localfile** from your machine to your home directory on the HPC, use `scp localfile username@hpc1.sun.ac.za:~/`. To copy a file **remotefile** from the HPC to your local machine in the current directory, use `scp username@hpc1.sun.ac.za:~/pathtofile/remotefile ..`

To copy an entire directory **localdir**, use `scp -r localdir username@hpc1.sun.ac.za:~/`.

This list of commands is not exhaustive, see the `scp` man pages, or google it.

To enable X11 forwarding, use the `-X` parameter with SSH:

```
$ ssh -X username@hpc1.sun.ac.za
```

```
$ ssh -X username@hpc2.sun.ac.za
```

Windows clients

Files are transferred with the Secure Copy Protocol (SCP). MobaXterm (<http://mobaxterm.mobatek.net/>) and WinSCP (<http://winscp.net/>) make this really easy.

Jobs are submitted from the console (also called command line). MobaXterm (<http://mobaxterm.mobatek.net/>), PuTTY (<http://www.chiark.greenend.org.uk/~sgtatham/putty/download.html>) and TeraTerm Pro (<http://en.sourceforge.jp/projects/ttssh2/releases/>) are SSH clients which will allow console access.

MobaXterm

Install from <http://mobaxterm.mobatek.net/> (or local copy (https://hpc1.sun.ac.za/MobaXterm_Setup_9.4.msi), possibly outdated). A very nice client that combines SSH client, SCP client and X11 server. Install this or a combination of the programs below.

MobaXterm sometimes has issues with transferring large amounts of data. Use WinSCP instead if you run into trouble.

PuTTY

Install from <http://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html> (or local copy (<http://support.sun.ac.za/Utilities/PuTTY>), only available on campus, possibly outdated).

TeraTerm Pro

Install from <https://osdn.net/projects/ttssh2/releases> (or local copy (<http://support.sun.ac.za/TeraTerm>), only available on campus, possibly outdated).

WinSCP

Install from <https://winscp.net/eng/download.php> (or local copy (<http://support.sun.ac.za/Utilities/WinScp>), only available on campus, possibly outdated).

Xming

Install from <http://sourceforge.net/projects/xming>.

Xming allows graphical programs to run on the cluster, but display on your own Windows machine. For it to work correctly, **X11 forwarding** must be enabled in your SSH client. In TeraTerm Pro enable it in the *Setup|SSH Forwarding...* menu (remember to *Setup|Save setup...* and restart TeraTerm Pro). In PuTTY, enable it in the *Connection|SSH|X11* menu on the left when connecting to the server. You do not need this if you have MobaXterm.

Once your SSH connection is open, check whether it works by seeing whether the `DISPLAY` variable is set:

```
[username@hpc1 ~]$ export | grep DISPLAY
declare -x DISPLAY="localhost:10.0"
```

Make sure Xming is running in the system tray, and then run a program in your SSH console which has a graphical frontend (like `xterm`):

```
[username@hpc1 ~]$ xterm
```

A graphical `xterm` console should open on your Windows desktop.

Please note that you do not need to install this if you use MobaXterm.

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